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MIL-B-23431A(AS)
AMENDMENT 1
17 NOVEMBER 1988

MILITARY SPECIFICATION

BEACON, RADAR, AN/APN-154(V)

This amendment forms a part of MIL-B-23431A(AS), dated 15 September 1967, and is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

This specification revision, including Amendement 1, is not to be used for future RT 763/APN-154 designs.

PAGE 1

1.1: Add the following to the end of the paragraph:

"It shall also transmit X-band pulses when triggered by the associated equipment (see 6.9) in the"All weather Carrier Landing System" (ACLS) mode of operation."

PAGE 2

1.3: Add the following configurations to the paragraph:

Type Designation	Brief Description	Applicable Paragraph
"RT-763B/APN-154(V)	High band receiver - side mount with elapsed time indicator (see Figure 4)	3.4.1 Table I
RT-763C/APN-154(V)	High band receiver - side mount with elapsed time indicator (see Figure 4)	3.4.1 Table I
RT-763D/APN-154(V)	High band receiver - side mount with elapsed time indicator (see Figure 4 and paragraph 1.3.1)	3.4.1 Table I
RT-763E/APN-154(V)	High band receiver - side mount with elapsed time indicator (see Figure 4 and paragraph 1.3.2)	3.4.1 Table I"

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PAGE 2

1.3: Add the following subparagraphs as follows:

"1.3.1 ACLS Operation - The RT-763D/APN-154(V) and the RT-763E/APN-154(V) configurations incorporate external trigger capability for ACLS operation.

1.3.2 Solid State Oscillator - The RT-763E/APN-154(V) configuration contains a solid state oscillator."

PAGE 2

1.4: Add the following configuration as follows:

Type Designation	Brief Description	Applicable Paragraph
"C-4419B/APN-154(V)	Control with ACLS test light and switch	3.4.6"

PAGE 2

1.6: Add the following configuration as follows:

Type Designation	Brief Description	Applicable Paragraph
"CG-2574A/APN-154(V)	Rf cable - 8 inches long	3.4.4"

PAGE 2

2.1: Add the following specification as follows:

"MIL-P-7788 Panels, Information, Integrally Illuminated"

PAGE 5

3.2.10: Change the primary input power requirements as follows:

"1.30 Amps"

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PAGE 6

After 3.3.2.1.2 add a new subparagraph as follows:

"3.3.2.1.3 High band Receiver - RT-763B/APN-154(V), RT-763C/APN-154(V), RT-763D/APN-154(V) and RT-763E/APN-154(V) shall tune from 9000 to 9600 MHz. Unless otherwise specified in the contract, the receiver shall be adjusted to 9310 plus or minus 2 MHz when shipped."

PAGE 6

3.3.2.2: Add the following sentence to the end of the paragraph:

"Unless otherwise specified in the contract, the transmitter frequency shall be adjusted to 9310 plus or minus 2 MHz when shipped."

PAGE 7

3.3.2.5: Add a new subparagraph as follows:

"3.3.2.5.1 Pulse Repitition Frequency (PRF) - For RT-763D/APN-154(V) and RT-763E/APN-154(V) the equipment shall be capable of operation at any prf from 10 to 4200 pps."

PAGE 7

3.3.2.11: Add a new subparagraph as follows:

"3.3.2.11.1 Power Output - For RT-763E/APN-154(V) the peak power output of the radar beacon shall be 300 Watts minimum under all conditions specified herein. Power output shall be measured at the antenna terminal of the duplexer-circulator."

PAGE 7

3.3.2.12: Add a new subparagraph as follows:

"3.3.2.13 Transmitter Short Term Stability - For RT-763E/APN-154(V), beginning ten seconds after interrogation starts and during the next five minutes, transmitter frequency shall not change more than 5 MHz."

PAGE 8

3.4.1: Change the number of configurations listed in line one as follows:

"twelve"

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PAGE 8

3.4.1: Add the following configurations to Table I:

Item	Form Factor	Maximum Weight (lbs)	Tuning Range	Mounting	Elapsed Time Indicator
"RT-763B/APN-154(V)	Fig. 4	4.4	9000-9600 MHz	Side	Yes
RT-763C/APN-154(V)	Fig. 4	4.4	9000-9600 MHz	Side	Yes
RT-763D/APN-154(V)	Fig. 4	4.4	9000-9600 MHz	Side	Yes
RT-763E/APN-154(V)	Fig. 4	4.4	9000-9600 MHz	Side	Yes"

PAGE 13

3.4.4: Add the following to the end of the paragraph:

"and CG-2574A/APN-154(V)"

PAGE 13

3.4.4.2: Add the following sentence to the end of the paragraph:

"For CG-2574A/APN-154(V) the length shall not exceed 8 inches."

PAGE 13

After subparagraph 3.4.5.2, add a new paragraph as follows:

"3.4.6 Control C-4419B/APN-154(V) - The control shall meet the applicable requirements of MIL-C-6781 for Type I control panels."

PAGE 13

Add the following subparagraphs after paragraph 3.4.6:

"3.4.6.1 Function - This unit shall provide controls for turning on the radar beacon system, for selecting the mode of decoder operation, and for selecting and testing the ACLS mode of operation when operated with the associated equipment listed in 6.9.

3.4.6.2 Form Factor - The control shall have a maximum allowed size of 5.75 inches width, 1.5 inches heigth and a 2 inch depth, excluding the integral interconnecting cable and front panel controls. Provisions shall be made for mounting on the aircraft cockpit console.

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3.4.6.3 Controls - The following controls shall be provided:

a. POWER - A three-position switch shall be provided for controlling power to the radar beacon system and, when activated, the ACLS system. The positions shall be labeled "OFF", "STBY" and "PWR". The PWR position shall enable input primary power to be applied to the radar beacon system. The ACLS position shall enable primary input power to be applied to both the radar beacon system and the augmentor/receiver R-1623/APN. The STBY position shall disable the decoder assembly. The OFF position shall disconnect primary input power.

b. MODE - A rotary selector switch shall be provided for selecting the mode of operation of the system. The switch, when in ACLS position, shall also provide for processing Ka-band interrogating signals in augmentor/receiver R-1623/APN, which provides external trigger for the RT-763/APN-154(V). Positions shall be as follows:

ACLS, SINGLE, DOUBLE-1, DOUBLE-2, DOUBLE-3, DOUBLE-4, DOUBLE-5.

c. ACLS TEST - A push-to-test switch with a dimming type indicator light shall be provided. When the MODE switch is in ACLS position, the ACLS TEST switch shall provide for illumination during normal ACLS receiver operation if interrogated by Ka-band signals. When processed, the switch shall provide for a test signal (BIT) for testing the augmentor/receiver when Ka-band interrogations are not present.

3.4.6.4 Panel - The front panel of the control shall be integrally illuminated in accordance with MIL-P-7788, Class 1-R, Type IV. Unless otherwise specified in the contract, illumination shall be operated from a 5 volt aircraft panel lighting circuit. The lighting shall be red unless otherwise specified in the contract.

3.4.6.5 Electrical Connections - Connections to external circuits shall be provided by means of an integral cable terminated in a standard connector in accordance with MIL-E-5400."

PAGE 20

6.4: Change the word "of" in line two to "or".

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PAGE 21

After paragraph 6.8, add the following new paragraph:

"6.9 Associated Equipment - The radar beacon should be capable of operating as part of the ACLS in conjunction with the following equipment:

a.	Modulator, Beacon	MD-817/APN
b.	Receiver, Radar	R-1623/APN"

As applicable, substitute

"Hz" for "cps",
"kHz" for "kc",
"MHz" for "mc",
"V" for "v",
"Vdc" for "vdc",
"dB" for "db",
"dBm" for "dbm" and
"us" for "usec"

in paragraphs 3.2.3, 3.2.3.1, 3.2.9, 3.2.11.2, 3.3.2.1, 3.3.2.1.2, 3.3.2.2, 3.3.2.3, 3.3.2.4.1, 3.3.2.4.2, 3.3.2.4.3, 3.3.2.6, 3.3.2.8, 3.3.2.9, 3.3.2.12, 3.4.11, 3.4.1.2.1, 3.4.1.2.2, 3.4.1.3.1, 3.4.1.3.2, 3.4.1.5, 3.4.1.8, 3.4.1.17, 3.4.3.1, 3.4.5.2, and 4.4.1.3.

Preparing Activity
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